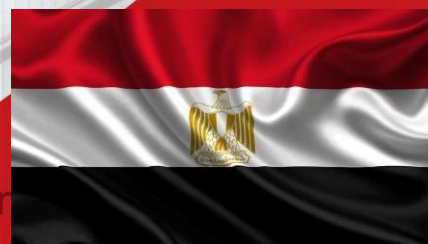
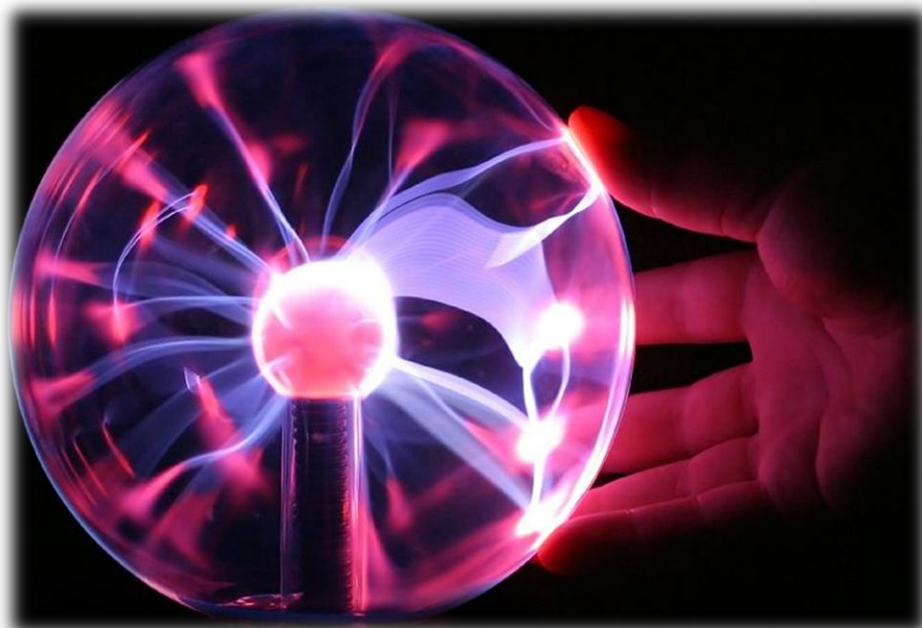


2023-2024

Physics 99 Lab.



Experiment NO: (1)

Ohm's law

Aim

- To verify the ohm's law and hence determine the unknown resistance of the given material of the wire.
- Verification of Series and parallel connections and compare between theoretical and experimental values. (Theoretical equals or nearly equal Experimental).

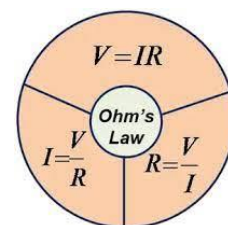
Apparatus



Theory

- The fundamental relationship among the three important electrical quantities **current**, **voltage**, and **resistance** was discovered by **Georg Simon Ohm**
- The **electric current** flowing through a conductor is **directly proportional** to the **potential difference** and **inversely proportional to resistance** across its ends provided the physical conditions (temperature, dimensions, pressure) of the conductor remains the same.

$$V = IR$$



where **R** is the **resistance** of the resistor in **Ohm (Ω)** when **potential difference (V)** is in **Volt** and **current (I)** in **Ampere (A)**. Resistance is a measure of how difficulty to flow current through the device.

- In this experiment you will **verify Ohm's law** in several different circuits using a **Ammeter** and a **Voltmeter**.
- You will also **apply Ohm's law** to determine the equivalent resistance of resistors connected in different combinations. Fig. a shows **the resistors connected in series** and Fig. b shows **the resistors connected in parallel**.



Fig. a. Series combination of resistors

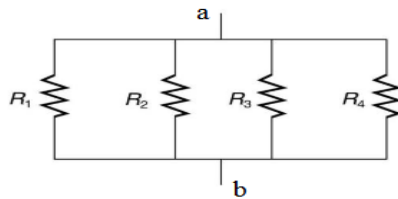


Fig. b. Parallel combination of resistor

- Basic circuit diagram for this lab is shown in Fig. C. A resistor (a and b are terminals of the resistor) is **connected** to a variable **power supply** to **change the current** through the resistor. In order to **measure the current** an **ammeter** is connected **in series** and to **measure potential difference** a **voltmeter** is connected **in parallel**.

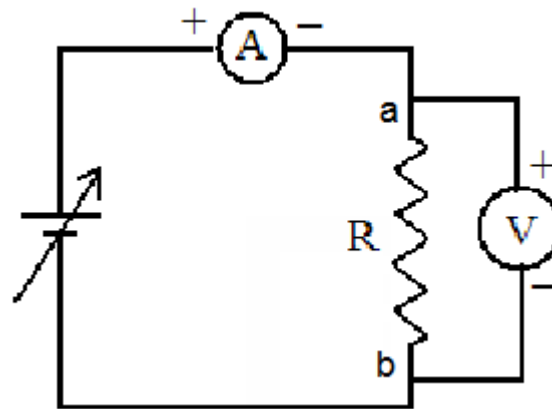


Fig. C. Basic circuit diagram for this lab

- Ammeters are connected in series so that the current flows through them.*
- Voltmeters are connected in parallel to resistive elements in circuit so that They measure the potential difference across the element.*
- Ideal voltmeter has infinite internal resistance.*

- Why are the ammeter connected in series and voltmeter in parallel to the resistor?

As we are using a variable power supply, voltage, and current sensors, in our real circuit. Ammeter is the current sensor and Voltmeter is the voltage sensor. Voltage sensor reads positive when the potential at the red (positive) terminal is higher than that at the black (negative) terminal.

General information

Consider Ohms law $V=IR$. what is the resistance of your skin: dry or wet.

If it is wet, is it water or sweat? if sweat, contains salt, salt water is good conductor.

The resistance different for different situations (very dry skin $R=500000\ \Omega$, skin wet with salt $R=100\ \Omega$ so we can calculate the current that will flow through your body if we know the voltage device.

Procedure

1. Connect circuit as show in fig (1)
2. Record the value of volt(V) and current(I) in table as show in table (1).
3. Draw relation between volt in (x-axis) and current in (y-axis) and obtain straight line Calculate the slope $=1/R$ (ohm) as show in Fig. (2).

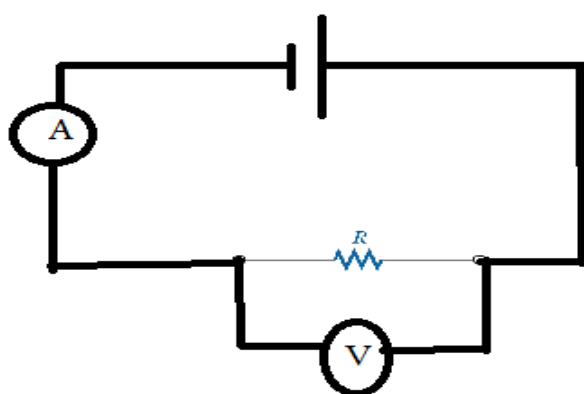


Fig. (1)

V (v)	I(mA)
2	
4	
8	

Table (1)

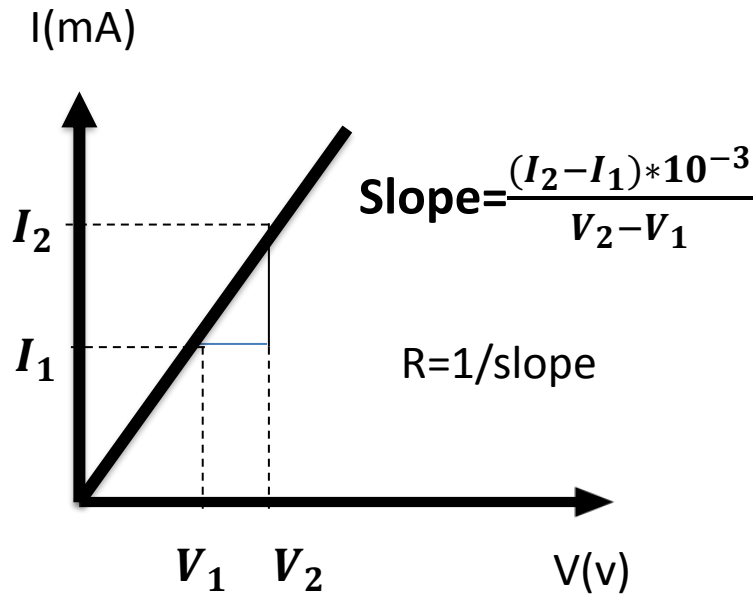


Fig. (2)

Series Connection

1. Connect circuit as show in Fig. (3).
2. First, R_1 , R_2 have known value Calculate (R_s) theoretical by $R_s = R_1 + R_2$
3. Second, switch circuit and record the values (V), (I) and calculate the resistance experimentally from the law $R = V/I$.
4. Compare between the values of R (theoretical, experimental)

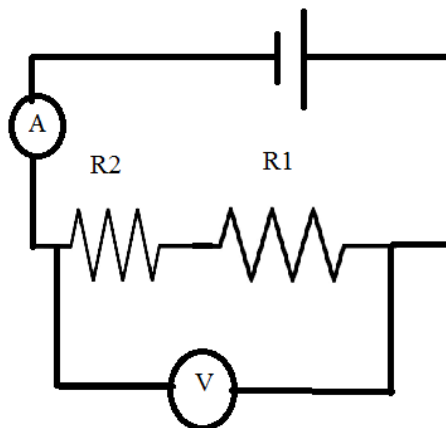


Fig. (3)

Parallel Connection

1. Connect circuit as show in Fig (4).
2. First, R_1 , R_2 have known value Calculate (R_p) theoretical by

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

3. Second, switch circuit and record the values (V), (I) and calculate the resistance experimentally from this law $R=V/I$.
4. Compare between the values of R (theoretical, experimental).

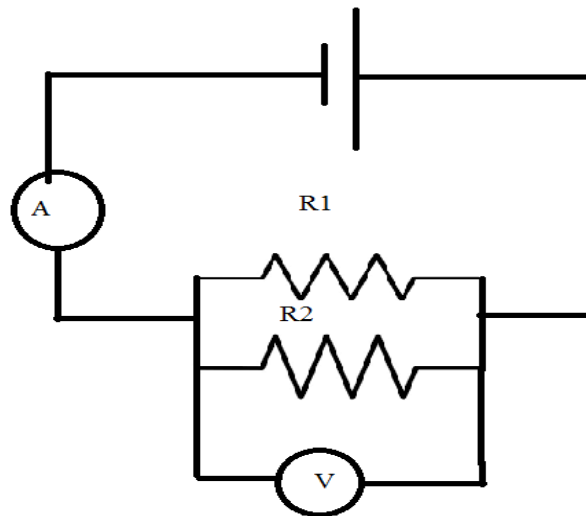


Fig. (4)

Results

V (v)	I(mA)
2	
4	
8	

Table (1)

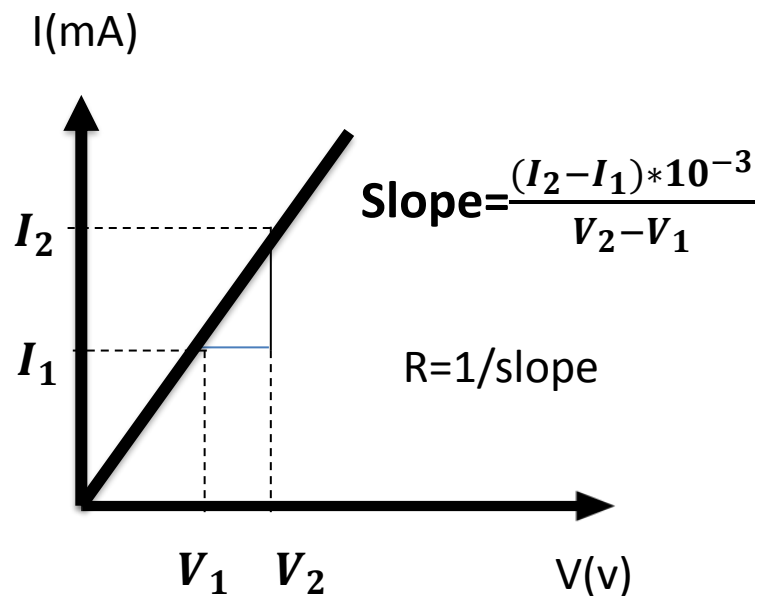


Fig. (3)

Part 2 (Series Connection)

Theoretical $R_s = (R_1 + R_2) \text{ K}\Omega$

Experimental $V = \dots \text{ volt}$ $I = \dots \text{ mA}$, $R = \frac{V}{I \times 10^{-3}} = \dots \Omega$

Should be Theoretical $(R_s) = \text{Experimental (R)}$

Part 3 (Parallel Connection)

Theoretical $\frac{1}{R_p} = \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \text{ K}\Omega$

Experimental $V = \dots \text{ volt}$ $I = \dots \text{ mA}$, $R = \frac{V}{I \times 10^{-3}} \Omega$

Should be Theoretical $(R_p) = \text{Experimental (R)}$

Note: To convert from Ω to $\text{K}\Omega$, You should multiply by 10^{-3}

